

Smoothed-LSTM: Advancing spatio-temporal NDVI prediction in semi-automated dataset for rice crop

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Abstract—Accurate Normalized Difference Vegetation Index (NDVI) forecasting is crucial for effective agricultural planning. However, a good prediction of the same requires sufficient data, but structured data is not available in the public domain or open-source community. Also, most of the existing methods do not consider spatial information. This study presents a novel semi-automated dataset generation framework that utilizes Sentinel-2, POWER Data Access Viewer, and Google Earth Engine to create a comprehensive time-series dataset. We propose a smoothed long short-term memory (LSTM) model considering time series, historical meteorological, and spatial information. The proposed Smoothed-LSTM-based model outperforms Traditional-LSTM models, demonstrating its effectiveness in NDVI prediction for agricultural applications.

Index Terms—NDVI, novel semi-automated framework, Sentinel-2, Smoothed-LSTM

I. INTRODUCTION

Predictive technologies in agriculture, utilizing data analytics and machine learning, offer farmers the potential to optimize resource usage, reduce waste, and increase profitability through informed decision-making. Additionally, early issue detection and proactive measures enhance risk mitigation, ensuring the long-term sustainability of agricultural operations. One such predictive variable is the Normalized Difference Vegetation Index (NDVI), used as a metric for assessing plant health and stress as shown by [1], [2].

The statistical methods such as AutoRegressive (AR), AutoRegressive Integrated Moving Average (ARIMA), Seasonal AutoRegressive Integrated Moving Average

(SARIMA), and other approaches for analyzing time series-based modeling of NDVI as used by [3], [4]. They have limitations in capturing specific complex patterns, which deep learning models better address in [5]. People have applied various machine learning (ML) algorithms [6], [7] for NDVI and EVI forecasting for different crops like wheat, rice, tomato, potato, and corn, etc. in [8]. Deep Learning (DL) algorithms autonomously learn complex features from raw data, boosting accuracy over traditional ML. In a comparative study, [9] proved that Long Short-Term Memory (LSTM) outperformed linear regression for time series prediction, emphasizing its superiority in handling sequential data. Also, the LSTM network effectively learns seasonal patterns and displays proficiency in capturing sudden changes [10]. There is a growing emphasis on incorporating spatio-temporal aspects in research [11]. In predicting NDVI in agriculture, researchers strive to emphasize spatial elements [8]. As discussed in large-area NDVI forecasting on both temporal and spatial cases, LSTM incorporates a multivariate, multi-step forecasting methodology, considering previous NDVI values and meteorological data as input.

DL architectures are sensitive to outliers, mitigated by filtering operations. Among finite impulse response (FIR) filters, the Blackman window exceeds others [12], [13], for effective smoothing.

Different datasets can be used for other purposes. Some researchers utilize a multispectral camera for their respective projects, typically mounted on an unmanned aerial vehicle (UAV) [14]. In contrast, others rely on remote sensing data obtained from MODIS, Sentinel-2, and Landsat 8 satellite

images [15]–[18]. Some of them [14] address the cost issue in crop monitoring. In contrast, [16] focusing on spatio-temporal data fusion based on very deep super-resolution (VDSR) to fuse the NDVI retrievals from Sentinel-2 and Landsat 8 images, and [17] present a spatio-temporal fusion algorithm for high-resolution vegetation monitoring and experimental results demonstrating that it outperforms existing algorithms like Enhanced Spatial and Temporal Adaptive Reflectance Fusion Model (ESTARFM) and Linear Mixing Growth Model (LMGM) in terms of accuracy and preservation of spatial details. However, in our specific case, we extend this approach by including the spatial information of each field alongside the previous NDVI values and meteorological data as additional input features.

In this paper, we propose that the Smoothed-LSTM outperforms the existing models regarding NDVI prediction for rice crops.

II. DATASET DESIGNING FOR SEMI-AUTOMATED FRAMEWORK

In this work, we use the data from the Sentinel-2 mission. Sentinel-2 is an Earth observation mission from Copernicus Programme that routinely gathers optical imagery at high spatial resolution ranging from 10m to 60m over land and coastal waterways. Sentinel-2A and Sentinel-2B are the two satellites that comprise the mission's current constellation. The Multispectral Instrument (MSI) Level-2A satellite collected data from multiple growing seasons between January 5, 2019, and June 28, 2021. We choose 505 agricultural fields in the Midnapore region in West Bengal, India, to prepare the time series dataset. We collect geometry information about fields and crops grown in them using Google Earth Engine, obtaining 88875 time-steps, which are individual observations over time. In addition to that, we gather meteorological data, including 2-meter temperature and Earth skin temp from the POWER Data Access Viewer. Our primary interest is in the rice produced in Midnapore, West Bengal. In Figure (1), each dot represents one field for which the time series data is prepared.

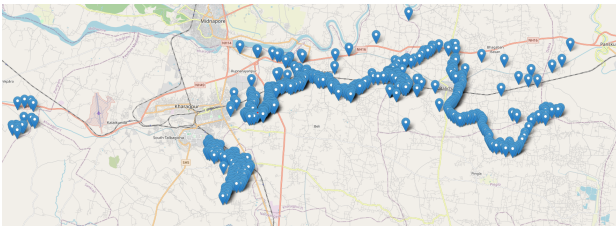


Fig. 1. The dataset was collected from Midnapore (West) from January 5, 2019 to June 28, 2021.

A. Normalized Difference Vegetation Index (NDVI)

The NDVI quantifies greenness on a scale from -1 to 1, where -1 indicates minimal greenness and 1 indicates

maximal greenness, as stated in Equation (1).

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (1)$$

Red and NIR are the surface reflectances of Red and near-infrared in different sensors. NDVI can be calculated using data from various sensors, including those on Landsat 8 (LC8 OLI/TIRS), Landsat 7 (LE7 ETM+), MODIS, and Sentinel satellites. There was an increase in NDVI throughout the growing season. However, as crop harvesting got underway, NDVI began to fall again. NDVI values are unique for each crop [19].

B. Exploratory Data Analysis

Initially, we had 'features' as variables or inputs (used in predictive models), such as Earth skin temperature and temperature at 2 meters, humidity at 2 Meter (g/kg), pressure on the surface, and wind speed at 10 meters as features. However, we find Earth skin temperature and temperature at 2 meters features have more impact based on their high information gain (quantifies the reduction in entropy achieved by selecting a particular feature for splitting data in ML algorithms) values of 0.916701 and 0.897144, respectively. Information gain is calculated by subtracting the weighted average of subset entropies from the entropy of the parent dataset, representing the reduction in disorder achieved by splitting data based on a feature; the feature with the highest information gain is selected in ML. We have used the scikit-learn library for this calculation. Studies have also revealed the impact of the above two features on NDVI [20]–[22].

C. Dataset Smoothing Process

Filters are indispensable tools in signal processing for enhancing or extracting valuable data insights. Techniques such as Rolling Mean, Gaussian smoothing, median filtering, Savitzky-Golay filtering, Exponential Smoothing, and FIR filtering offer diverse noise reduction methods. The Blackman FIR filter stands out for its smooth frequency response and low passband ripple. It is particularly suitable for applications requiring balanced frequency selectivity and time-domain response, such as audio or biomedical signal processing. Nonetheless, longer filter lengths may increase computational demands and potential phase distortion, necessitating meticulous parameter tuning for optimal performance. After evaluating several filters, including Rolling Mean, Savitzky-Golay, and FIR filters, we determined the Blackman FIR filter to be the most effective for data smoothing, as validated through experiments [12], [13] detailed in Equation (2).

$$h[n] = w[n] \cdot \left(\frac{\sin(2\pi f_c(n - M/2))}{\pi(n - M/2)} \right) \quad (2)$$

Where $h[n]$ is the filter impulse response, $w[n]$ is the Blackman window function applied to the filter coefficients, f_c is the normalized cutoff frequency, M is the filter order, and n is the discrete-time index.

The Blackman window function is defined as Equation (3):

$$w[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{M}\right) + 0.08 \cos\left(\frac{4\pi n}{M}\right) \quad (3)$$

III. METHODOLOGY FOR RNN-BASED ARCHITECTURE

Traditional-LSTM models are vulnerable to outliers, but smoothing techniques can mitigate this issue, resulting in improved model predictability. In our proposed approach, detailed in Table (I), we utilize a stacked Smoothed-LSTM architecture, which ensures stability and reproducibility by fixing the random seed and maintaining consistent training conditions. Since the initial model parameters depend on the random distribution, setting the random seeds stabilizes these parameters, promoting consistent training outcomes and facilitating result reproducibility.

To address LSTM's sensitivity to input scale, MinMaxScaler is employed for preprocessing, with data divided into 80%

Our stacked Smoothed-LSTM model enhances predictive accuracy, utilizing default Sigmoid activation and multivariate time series data analysis for NDVI forecasting focusing on the previous k time points. The 'return_sequences' option optimizes information flow between LSTM layers. For 100 epochs, we trained the network with a batch size of 8, specifying the number of data samples processed before updating the model's parameters. A dropout rate of 0.2 refers to the proportion of randomly selected neurons to ignore during training, helping prevent overfitting. We optimized its performance using the Adam optimizer.

TABLE I
MODEL SUMMARY OF SMOOTHED-LSTM

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 5, 180)	133920
lstm_1 (LSTM)	(None, 80)	83520
dropout (Dropout)	(None, 80)	0
dense (Dense)	(None, 1)	81
Total params & Trainable params: 217521 (849.69 KB)		
Non-trainable params: 0 (0.00 Byte)		

IV. EXPERIMENTAL RESULTS

We evaluate various error metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE), with a preference for RMSE due to its effective outlier penalization and capability for original unit comparison [23]. Our comprehensive analysis, detailed in Table (II), demonstrates that Smoothed-LSTM outperforms not only neural network models like Traditional-LSTM but also statistical methods such as AR, ARIMA, and SARIMA for NDVI time series prediction. Although Traditional-LSTM lacks dataset smoothing filters, a comparative table illustrates that employing

Rolling Mean, Savitzky-Golay, and FIR filters significantly enhances predictive performance. Additionally, our findings suggest that the proposed Blackman FIR filter, incorporated within the Smoothed-LSTM model, yields the most accurate predictions among the evaluated methods.

TABLE II
COMPARING THE SMOOTHED-LSTM MODEL WITH EXISTING MODELS

Model	Train RMSE	Test RMSE
AR	7.38e-02	8.43e-02
ARIMA	8.33e-02	8.75e-02
SARIMA	9.28e-02	9.98e-02
Traditional-LSTM	4.66e-02	1.33e-01
Savitzky-Golay	1.95e-02	6.34e-02
Rolling Mean	1.92e-02	6.21e-02
Smoothed-LSTM	2.49e-03	3.88e-03

Figure (2) illustrates the enhanced performance of the Smoothed-LSTM compared to the Traditional-LSTM. Both figures (2(a)) & (2(d)) clearly show that the Smoothed-LSTM with Blackman FIR filter follows the data trend better than the Traditional-LSTM. Also, both figures (2(b)) & (2(c)) are not performing well with respect (2(d)).

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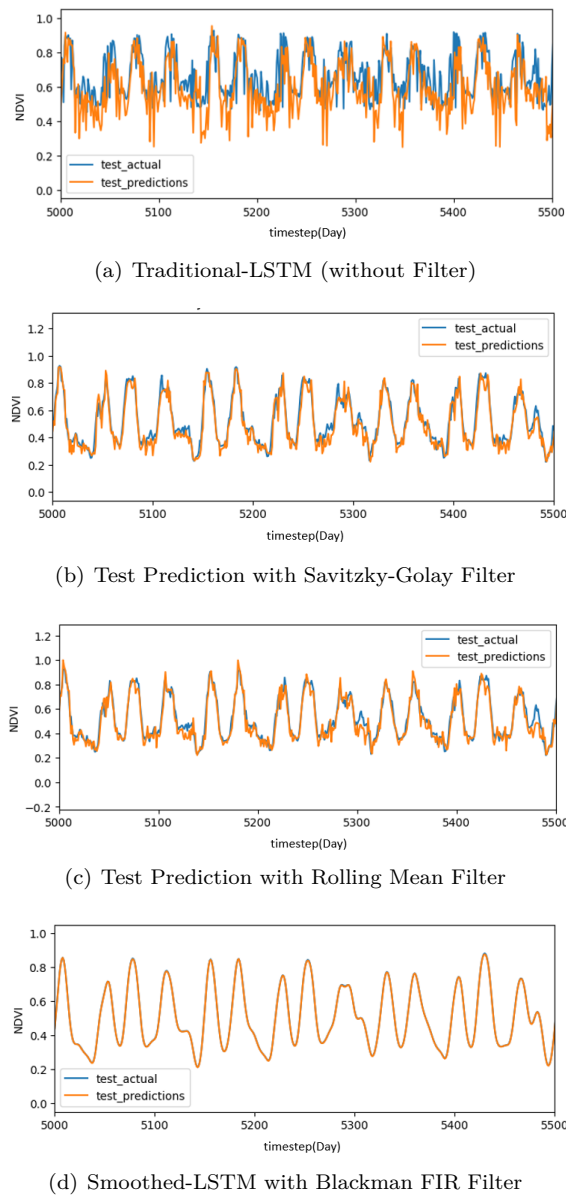


Fig. 2. Test Prediction for Traditional-LSTM vs Smoothed-LSTM along with Savitzky-Golay, Rolling Mean, Blackman FIR Filters

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